

- ❖ Write the characteristics of GSM and its disadvantages. What has changed, what is the same if you compare GSM with today's systems?

Characteristics:

- Communication: mobile, wireless communication; support for voice and data services
- Total mobility: international access, chip-card enables use of access points of different providers
- Worldwide connectivity: one number, the network handles localization
- High capacity: better frequency efficiency, smaller cells, more customers per cell
- High transmission quality: high audio quality and reliability for wireless, uninterrupted phone calls at higher speeds (e.g., from cars, trains)
- Security functions: access control, authentication via chip-card and PIN

Disadvantages:

- There is no perfect system!!
- no end-to-end encryption of user data
- no full ISDN bandwidth of 64 kbit/s to the user, no transparent B-channel
- reduced concentration while driving
- electromagnetic radiation
- abuse of private data possible

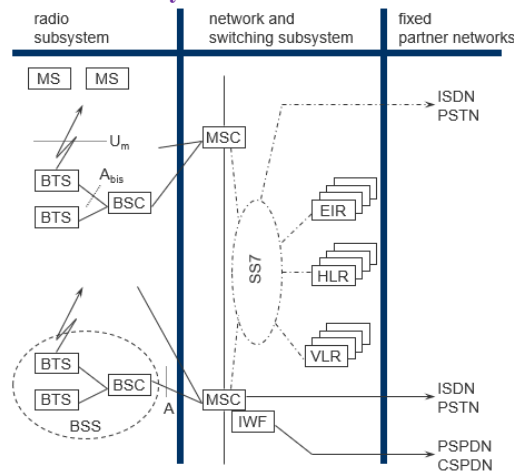
What has changed:

- Data speed: GSM (up to 50 kbps) → Today's systems (4G: 100+ Mbps, 5G: Gbps).
- Coverage & capacity: Modern systems support higher user density and global coverage.
- Services: GSM mainly voice & SMS → Today's systems support video calls, streaming, IoT, and high-speed internet.

What is the same:

- Core goal: mobile communication for voice and data.
- Cellular network structure with base stations and mobile switching centers.
- Subscriber identification using SIM cards (evolved but still used).

- ❖ Draw the architecture of the classical GSM system and describe the functionality of the components.



The classical GSM system is divided into three main parts: the **Mobile Station (MS)**, the **Base Station Subsystem (BSS)**, and the **Network and Switching Subsystem (NSS)**. The MS, consisting of the mobile device and SIM card, provides communication through the air interface and stores user identity and authentication data. The BSS, made up of the **Base Transceiver Station (BTS)** and **Base Station Controller (BSC)**, manages radio communication and controls multiple BTSs, handling channel allocation and handovers. The NSS includes key components like the **Mobile Switching Center (MSC)** for call control and routing, the **Home Location Register (HLR)** and **Visitor Location Register (VLR)** for storing subscriber information, the **Authentication Center (AUC)** for generating security keys, and the **Equipment Identity Register (EIR)** for verifying device identity. Together, these components enable secure, reliable voice and data communication in GSM networks.

❖ **Why does GSM separate the MS and SIM?**

SIM stores subscriber identity securely; MS handles device hardware. Allows portability across devices.

❖ **What is the advantage/disadvantage of an eSIM?**

- Advantage: Remote provisioning, no physical SIM swap needed.
- Disadvantage: Harder to remove/replace manually, depends on carrier support.

❖ **Go once more through the channel/slot structure and understand where GSM applies what type of multiplexing!**

- FDM (Frequency Division Multiplexing): divides spectrum into 200 kHz carrier channels.
- TDMA (Time Division Multiple Access): each carrier is divided into 8 time slots, each serving one user per frame.

❖ **How does GSM adapt to varying distances between MS and BTS? What is the role of the guard spaces?**

- Timing Advance (TA): adjusts MS transmission to align with BTS timing.
- Guard spaces: prevent time slot overlap caused by propagation delay.

❖ **What limits the radius of a GSM cell?**

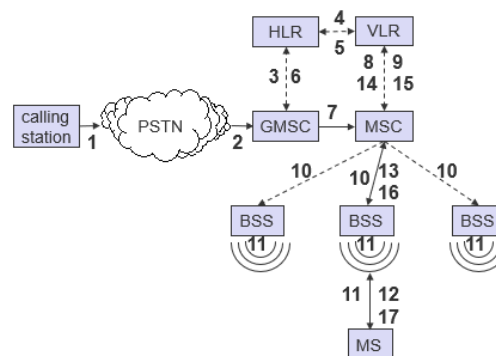
- Maximum ~35 km due to TDMA timing limits.
- Also limited by signal strength, interference, and propagation delay.

❖ **What is the maximum data rate per time slot? Why is the real data rate offered that much lower?**

- Maximum: 22.8 kbps per slot.
- Real rate: ~9.6–14.4 kbps due to error correction, signaling, and channel coding overhead.

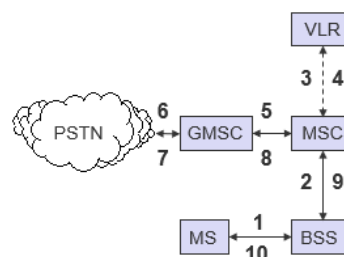
❖ **Draw the flowchart of Mobile Terminated Call (MTC).**

- 1: calling a GSM subscriber
- 2: forwarding call to GMSC
- 3: signal call setup to HLR
- 4, 5: request MSRN from VLR
- 6: forward responsible MSC to GMSC
- 7: forward call to current MSC
- 8, 9: get current status of MS
- 10, 11: paging of MS
- 12, 13: MS answers
- 14, 15: security checks
- 16, 17: set up connection



❖ **Draw the flowchart of Mobile Originated Call (MOC).**

- 1, 2: connection request
- 3, 4: security check
- 5-8: check resources (free circuit)
- 9-10: set up call



❖ **How does the GSM system locate an MS? How precise is the localization?**

- GSM locates a mobile using the cell ID of the serving base station.
- Precision can be improved with Timing Advance (TA) or triangulation using multiple BTS.
- Accuracy: typically 100–500 meters in urban areas, up to a few kilometers in rural areas.

- ❖ **What are the key mechanisms and algorithms used for data encryption in GSM, and how do they contribute to ensuring the security and privacy of communications in a GSM network?**

Key Mechanisms / Algorithms:

- A3 algorithm: Used for authentication of the subscriber. Confirms that the MS is valid on the network.
- A8 algorithm: Generates session keys (Kc) from the subscriber key (Ki) for encryption.
- A5 family (A5/0, A5/1, A5/2, A5/3): Encrypts voice and data over the air interface between MS and BTS.

Contribution to Security and Privacy:

- Subscriber authentication (A3): Prevents unauthorized access to the network.
- Session key generation (A8): Ensures that encryption keys are unique per session, preventing eavesdropping.
- Air interface encryption (A5): Protects voice and data confidentiality, making intercepted signals unreadable.

- ❖ **Describe the functions of the MS and SIM. Why does GSM separate the MS and SIM? How and where is user-related data represented/stored in the GSM system? How is user data protected from unauthorized access, especially over the air interface? How could the position of an MS (not only the current BTS) be localized? Think of the MS reports regarding signal quality.**

Describe the functions of the MS and SIM. Why does GSM separate them?

- MS (Mobile Station): Handles radio transmission/reception, signal processing, and user interface (voice, SMS, data).
- SIM (Subscriber Identity Module): Stores subscriber identity (IMSI), authentication keys (Ki), and security info.
- Reason for separation: Allows user portability—the SIM can be moved to another device while keeping the subscriber profile intact.

How and where is user-related data represented/stored in the GSM system?

- ✓ HLR (Home Location Register): Permanent subscriber info and service subscriptions.
- ✓ VLR (Visitor Location Register): Temporary info while roaming.
- ✓ AUC (Authentication Center): Stores encryption keys for secure communication.
- ✓ EIR (Equipment Identity Register): Stores device IMEI for validity.

How is user data protected from unauthorized access, especially over the air interface?

- ✓ Authentication (A3 algorithm): Confirms subscriber identity.
- ✓ Session key generation (A8): Creates unique encryption keys per session.
- ✓ Air interface encryption (A5): Encrypts voice and data between MS and BTS.
- ✓ SIM security: Stores sensitive keys securely; not accessible by the MS directly.

How could the position of an MS (not only the current BTS) be localized?

- ✓ Cell ID: Basic location via serving BTS.
- ✓ Timing Advance (TA): Estimates distance from serving BTS.
- ✓ Neighbor cell measurements: MS reports signal strength/quality from neighboring BTS.
- ✓ Triangulation / network-assisted localization: Uses multiple BTS measurements to improve accuracy.
- ✓ What is protected by the GSM security mechanisms?

- ❖ **What is missing compared to today's communication systems?**

End-to-end encryption, strong mutual authentication, and protection against modern attacks like IMSI catchers.

- ❖ **What multiplexing schemes are used in GSM for what purposes? Think also of other layers apart from the physical layer.**

Physical Layer:

- FDM (Frequency Division Multiplexing): divides the spectrum into 200 kHz carrier channels.
- TDMA (Time Division Multiple Access): each carrier has 8 time slots to serve multiple users.

Data Link / Network Layers:

- Statistical Multiplexing: used for packet-switched services (e.g., GPRS) to share resources efficiently.
- Logical Channels Multiplexing: multiple logical channels (control, voice, SMS) are mapped onto physical time slots.

- ❖ **Why use a separated, specialized system like TETRA instead of the normal cellular network?**

Reasons for TETRA:

- Very fast call setup for emergency services.
- Group calls and push-to-talk functionality.
- High reliability and robustness in harsh environments.
- Secure communication with encryption.

Reasons against a separate system:

- Requires dedicated infrastructure, increasing cost.
- Cellular networks can offer voice and data services with some extras.
- Maintenance of two separate networks can be more complex.

- ❖ **Compare the handovers in GSM with the ones in UMTS. Why are they called hard or soft, respectively?**

- **GSM:**

- **Hard handover** – “break-before-make.” Connection to the current cell is released before connecting to the new cell.
- Reason: GSM uses **TDMA/FDM**, so a time slot cannot be shared simultaneously between cells.

- **UMTS:**

- **Soft handover** – “make-before-break.” Connection to the new cell is established **before** releasing the old one.
- Reason: UMTS uses **CDMA**, allowing the mobile to communicate with multiple cells at the same time.

- ❖ **Why did most GSM operators pick UMTS as 3rd generation and cdmaOne operators pick cdma2000?**

GSM → UMTS:

- GSM operators wanted backward compatibility with 2G networks.
- UMTS (W-CDMA) allowed a smooth migration path, reusing spectrum and core network concepts.

cdmaOne → cdma2000:

- cdmaOne operators preferred evolution within the CDMA family.
- cdma2000 maintained compatibility with existing CDMA networks and devices.

Reason: Operators chose the 3G standard that best preserved investment and allowed network continuity.

- ❖ **While the data rates increased over time, the delay remained more or less constant at about 100–150 ms from a mobile device to the base station. What problem arises if, e.g., downloading content using TCP via these fast connections?**

- TCP uses Round-Trip Time (RTT) for congestion control and flow control.
- Even with high data rates, a constant 100–150 ms delay limits the speed at which TCP can acknowledge packets, slowing down throughput.
- Problem: TCP cannot fully utilize the high bandwidth, leading to suboptimal download speeds despite fast connections.

❖ **Name high-level differences between LTE and GSM/GPRS/UMTS! What do they have in common?**

Feature	GSM / GPRS / UMTS	LTE	Common Points
Generation	2G / 2.5G / 3G	4G	Cellular networks
Access	TDMA/FDM (GSM), WCDMA (UMTS)	OFDMA (DL), SC-FDMA (UL)	Use radio spectrum to connect MS
Data Rate	Up to ~2 Mbps	Up to hundreds of Mbps	Voice & data services
Network	CS for voice, PS for data	Fully packet-switched	Base stations & core network
Handover	GSM: Hard, UMTS: Soft	Mostly hard	Mobility management
Latency	~100–150 ms	~10–30 ms	Signaling & scheduling
Services	Voice, SMS, MMS, low-speed data	High-speed data, VoLTE, streaming	Subscriber authentication & security

❖ **Compare the architecture of LTE / UMTS / GSM – what is new, what is basically the same?**

- New in LTE: Fully packet-switched network, simplified architecture, eNodeB integrates radio and control functions, no separate RNC.
- Same: Core concepts like MS, base stations, mobility management, subscriber authentication remain.
- UMTS vs GSM: UMTS adds WCDMA, RNC for control; GSM uses TDMA/FDM and MSC/VLR for circuit-switched calls.

❖ **Compare multiplexing / multiple access in LTE with UMTS / GSM – differences / similarities**

Layer	GSM	UMTS	LTE	Similarity
Multiple Access	TDMA/FDM	WCDMA	OFDMA (DL), SC-FDMA (UL)	All divide resources to serve multiple users
Multiplexing	Time slots per carrier	Spreading codes	Subcarriers + time slots	All map multiple logical channels to physical resources
Purpose	Share frequency channels	Share spectrum with codes	Efficient spectrum usage, high data rates	Enable multiple users to share the network

Exercises.

1. Name some key features of GSM, DECT, TETRA, and UMTS. Which features do the systems have in common? Why have the three older systems been specified? In what scenarios could one system replace another? What are the specific advantages of each system?

- **GSM:** Wide area coverage, 9.6–50 kbit/s, supports voice, SMS, MMS.
- **DECT:** Local coverage, supports voice and data, high user density.
- **TETRA:** Regional coverage, ad-hoc mode, very fast connection setup, group calls, voice and data, robust.
- **UMTS:** Medium coverage, higher data rates (384 kbit/s), flexible bandwidth.
- **Common features:** Circuit-switched voice, integrated into fixed networks, ISDN core network.
- **Reasons for older systems:** Optimized for specific scenarios (local, regional, high density).
- **Replacement:** GSM can replace DECT if licensing allows; modified GSM can replace TETRA (except ad-hoc mode).
- **Advantages:** GSM—wide coverage, TETRA—fast setup & ad-hoc, DECT—high density, UMTS—higher data rates.

2. What are the main problems when transmitting data using wireless systems made for voice? How can these problems be mitigated and efficiency improved? How can billing support this?

- **Problems:** Fixed bandwidth for maximum expected rate, long connection setup, delays in spontaneous data transmission.
- **Mitigation:** Introduce packet-switched services (like GPRS).
- **Billing:** Volume-based billing; application-based billing is better since users care about apps, not bytes.

3. Which types of different services does GSM offer? Why are these services separated? Give examples.

- **Bearer services:** Provide communication channels for data/voice.
- **Tele services:** Voice, SMS, MMS.
- **Supplemental services:** Call forwarding, call waiting, call barring.
- **Reason for separation:** Phased introduction, clear roles for network providers, service providers, and device manufacturers.

4. Why is the data rate available to a user lower than the maximum air interface rate in GSM?

- Forward error correction, signaling overhead, guard spaces, limited number of channels.

5. Name the main elements of GSM system architecture and describe their functions. What are the advantages of specifying internal interfaces?

- **MS (Mobile Station):** Device functions—radio, coder/decoder, ID.
- **BSS (Base Station Subsystem):** Connects MS to network, handles radio.
- **NSS (Network & Switching Subsystem):** Core network—switching, HLR, VLR.
- **Advantage of internal interfaces:** Multi-vendor support, flexibility, equipment from different vendors can interoperate.

6. Describe the functions of the MS and SIM. Why does GSM separate MS and SIM? Where is user-related data stored? How is it protected? How can the MS location be determined?

- **MS:** Device-related functions (radio, coders).
- **SIM:** Subscriber info, PIN, authentication.
- **Separation:** Users can change phones while keeping personal data.
- **Data storage:** SIM, VLR (temporary location), HLR (home network).
- **Protection:** Encryption over air, PIN protection, AuC-secured authentication.
- **Localization:** Signal strength from surrounding base stations, timing info.

7. How does the HLR/VLR architecture limit scalability in terms of users, especially moving users?

- VLR can store ~1 million users.
- Frequent movement causes high update load on HLR, reducing scalability.

8. Why is new infrastructure needed for GPRS but not for HSCSD? Which new components are introduced and their purpose?

- **HSCSD:** Uses existing GSM core.
- **GPRS:** Requires SGSN & GGSN routers (IP routing), user context, subscriber registry.

9. What are the limitations of a GSM cell in terms of diameter and capacity (voice/data) for GSM, HSCSD, and GPRS? How can capacity be increased?

- **Diameter:** ~70 km (timing limit).
- **Capacity:** Limited by channels $\times$ time slots.
- **Increase:** EDGE modulation, more slots, advanced coding.

10. What multiplexing schemes are used in GSM and their purposes?

- **SDM:** Cell layout, frequency reuse.
- **FDM:** Assign channels/frequencies to operators and cells.
- **TDM:** Assign time-slots to terminals for transmission.

11. How is synchronization achieved in GSM and why is it important?

- Terminals synchronize to frame structure; midamble improves timing.
- Ensures correct transmission and prevents data overlap.

12. What are the reasons for delays in GSM for packet data traffic? Compare circuit-switched and packet-switched.

- **Circuit-switched:** Connection setup (seconds), FEC (~100 ms), propagation.
- **Packet-switched:** Channel access (depends on load), FEC, routing delays.

13. Where and when can collisions occur in GSM? Compare GSM, HSCSD, GPRS.

- During connection setup (slotted Aloha).
- After slot allocation, no collisions occur in data transmission for GSM, HSCSD, or GPRS.

14. Why and when are different signaling channels needed in GSM?

- **SDCCH:** Setup & authentication before TCH.
- **SACCH:** Monitor signal quality during TCH.
- **FACCH:** Extra signaling using TCH slots during ongoing calls.

15. How is localization, location update, and roaming handled in GSM?

- VLR stores current location; HLR tracks which VLR.
- Roaming occurs via agreements (national/international).
- Paging done during call setup.

16. Why are so many identifiers/addresses needed in GSM? Distinguish user vs system identifiers.

- **User:** MSISDN (phone number).
- **System:** IMSI (subscriber ID), IMEI (device ID), TMSI/LMSI (temporary), MSRN (roaming number), CI/BSIC (cell/base IDs).
- Multiple IDs ensure privacy, location management, and system operations.

17. Reasons for handover in GSM and associated problems. Steps and resources for HSCSD/GPRS.

- Weak signal, high load.
- Allocate new channels/time-slots.
- HSCSD: needs same number of slots; GPRS: slots depend on load.
- QoS not guaranteed if new cell busy.

18. Functions of authentication and encryption in GSM. How is security maintained?

- Verify user and SIM.
- Use challenge-response scheme with AuC.
- Encrypt over air interface only.
- UMTS introduces full component authentication.

19. How can higher data rates be achieved in GSM? Differences for HSCSD, GPRS, EDGE?

- **HSCSD:** Combines multiple slots.
- **GPRS:** Packet switching, multiple slots, different coding.
- **EDGE:** New modulation scheme.
- Complexity increases; handover delays and coding delays remain.

20. What limits real device data rates for HSCSD/GPRS compared to theoretical GSM?

- Cannot use all 8 slots simultaneously.
- Switching delay from send/receive mode.
- Not all coding schemes supported.

21. Using 115.2 kbit/s GPRS with 0.5 s delay, how many bytes are in transit? How would TCP behave for a 10 kB transfer?

- $115.2 \text{ kbit/s} \times 0.5 \times 2 = 14.4 \text{ kB}$ in transit.
- TCP performs poorly for 10 kB (inefficient windowing, high latency).
- Needs adjustments like larger initial sending window.

22. How much of GSM network is used by GPRS? Which elements handle data transfer?

- Uses GSM core for localization and authentication.
- Data transfer done by SGSN and GGSN; MSC not needed.

23. What are typical DECT data rates? How are they achieved? Compare with GSM.

- 120 channels, 32 kbit/s each.
- TDMA (24 slots/frame), FDM (multiple channels), SDM (base station placement).
- Simpler than GSM, fewer handovers, simpler accounting.

24. Who uses trunked radio systems? Why are they attractive and cheaper than GSM?

- Police, fire, ambulance, disaster teams, taxis.
- Fast setup, group calls, ad-hoc, robust, reliable.
- Cheaper due to fewer base stations, less complex billing/accounting.

25. Summarize main features of 3G/UMTS. How are higher capacities and rates achieved?

- Higher data rates, better voice codecs, CDMA, 2 GHz operation.
- Asymmetrical rates via spreading factors.
- More capacity via powerful modulation, compression, and CDMA.

26. Compare mobile networks in Europe, Japan, China, North America.

- **Europe:** UMTS FDD, started 2002.
- **Japan:** FOMA (UMTS), cdma2000.
- **China:** UMTS, TD-SCDMA, some cdma2000.
- **North America:** EDGE or cdma2000; multiple systems, no single winner.
- No global standard yet.

27. What disadvantage does OVFSF have for flexible rates? How does UMTS offer different rates (FDD/TDD)?

- OVFSF: only fixed multiples of 15 kbit/s.
- **FDD:** Adjust spreading factor.
- **TDD:** Request slots for uplink/downlink separately.

28. How are different DPDCH channels distinguished in UTRA FDD and TDD?

- **FDD:** Same spreading codes; each UE has unique scrambling code.
- **TDD:** Cell-specific scrambling codes.

29. How are data streams combined/split in CDMA networks? What about soft handover?

- Done at NodeB or RNC, not MSC.
- Iur interface transfers data between RNCs.
- Soft handover possible using rake receivers and multipath combining.

30. Explain near-far effect in UTRA-FDD and GSM. How is it mitigated?

- **Near-far effect:** Stronger signals from nearby users overpower weaker ones.
- **UTRA-FDD:** Power control 1500×/sec ensures equal signal at base.
- **GSM:** TDMA/FDM separates users, avoiding near-far effect.